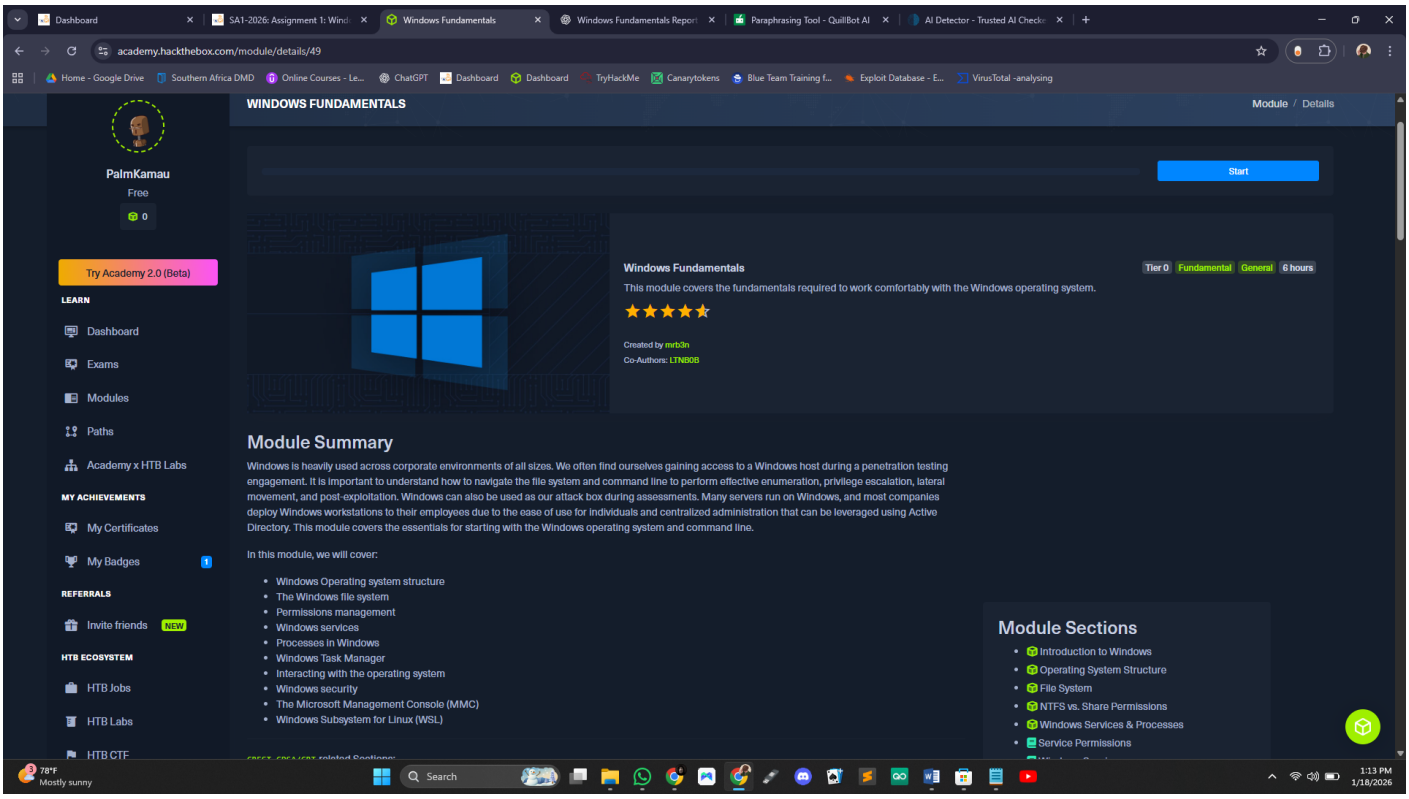
GMAIL: palmkamauk@gmail.com

REPORT: WINDOWS FUNDAMENTALS

Here is the link to my completed module:
https://academy.hackthebox.com/achievement/2272384/49

1.1 INTRODUCTION

I began the Windows Fundamentals session on Hack the Box (HTB) by learning how important Windows systems are for security assessments and in the real world of business I learned about Windows during the module and how it is often used in penetration testing. I also learned how it can be utilized as a target and an attack platform. The hands-on exercises taught me how to use and interact with Windows computers in a way that helped me study and do well in the HTB Academy.

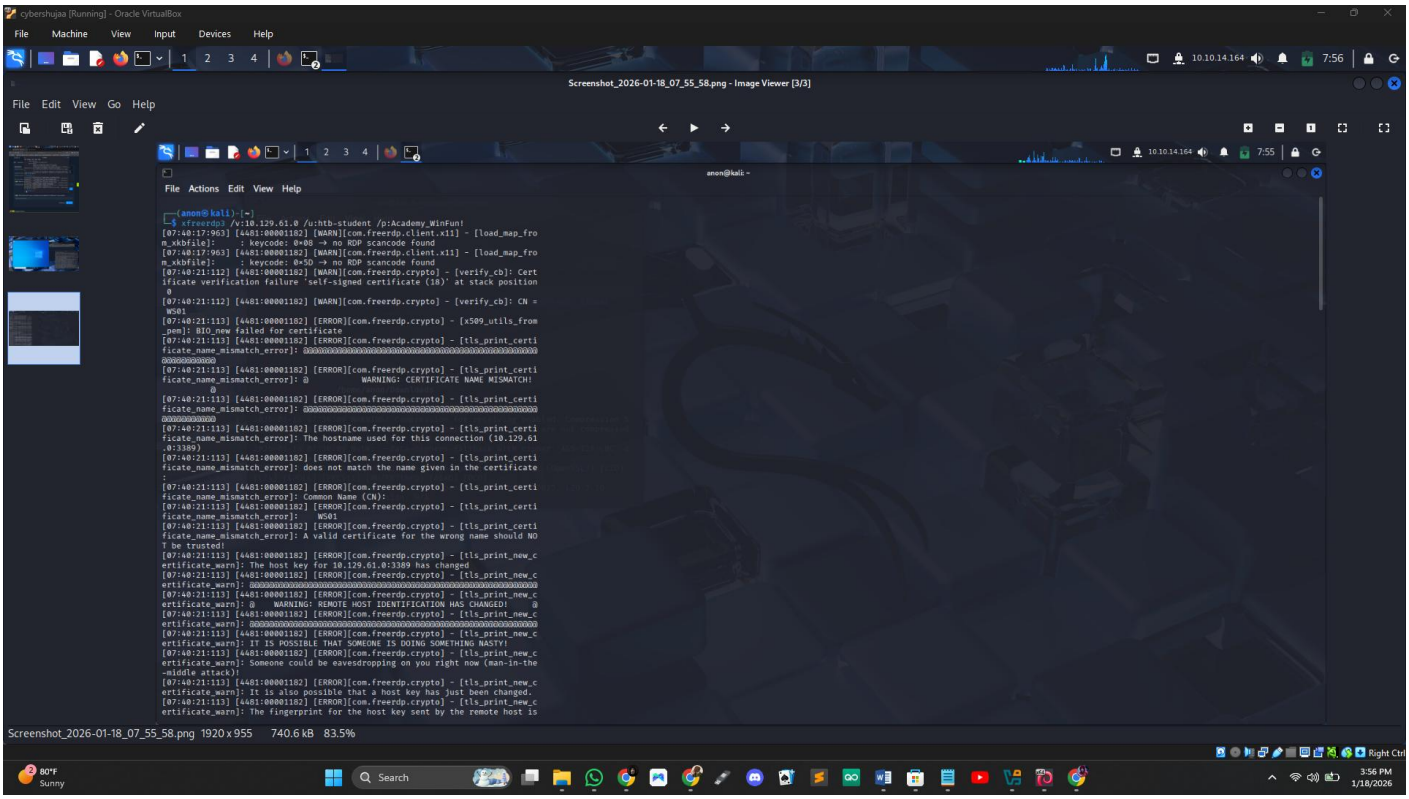


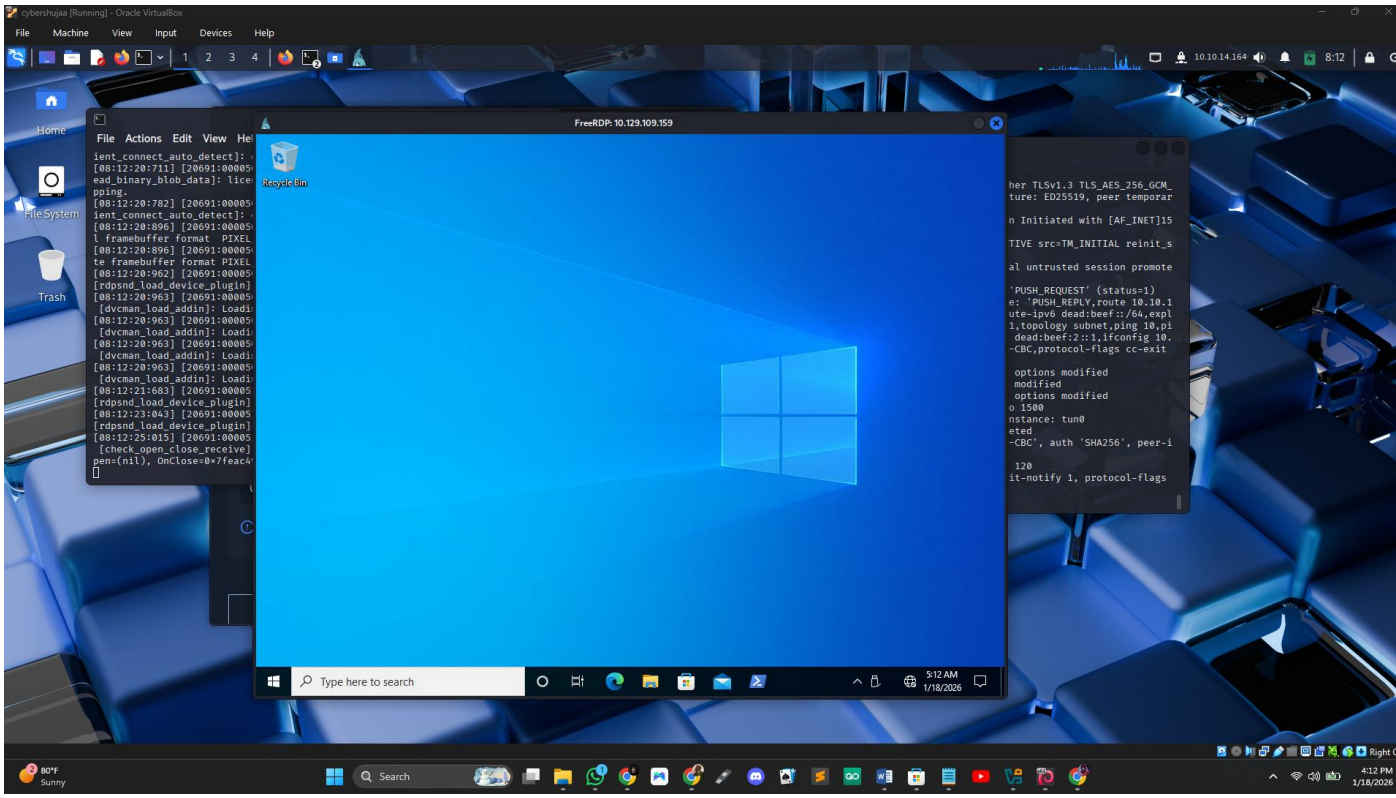
1.2 Connecting Remotely to Target

I then went ahead and learned about connecting to a windows machine remotely, both in a linux and also via windows distors, for the Linux distro, Kali linux, I used RPD to connect to the target windows with the help of xfreerpd3. The command "xfreerdp /v:<targetIp> /u:htb-student /p:Password" failed in my machine when I attempted to do the connection so I alternatively did my own research and found out that my kali version only accepted *xfreerpd3*, hence had to change my command to " xfreerdp3 /v:<targetIp> /u:htb-student /p:Password", where the v is for the target windows, u- for the username of the target Windows, p- password

for the username of the selected target window. But in our case, Hack The Box provided the target, username and the password.

The screenshot below portrays the process and command used in connecting to target Windows;


cybershujaa [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Screenshot_2026-01-18_07_55_58.png - Image Viewer [3/3]
File Edit View Go Help
anon@kali: ~
File Actions Edit View Help
┌──(anon㉿kali)-[~]
└─$ xfreerdp3 /v:10.129.61.0 /u:htb-student /p:Academy_WinFun!
[07:40:17:963] [4481:00001182] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x00 → no RDP scancode found
[07:40:17:963] [4481:00001182] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x5D → no RDP scancode found
[07:40:21:112] [4481:00001182] [WARN][com.freerdp.crypto] - [verify_cb]: Certificate verification failure 'self-signed certificate (18)' at stack position 0
[07:40:21:112] [4481:00001182] [WARN][com.freerdp.crypto] - [verify_cb]: CN = WS01
[07:40:21:113] [4481:00001182] [ERROR][com.freerdp.crypto] - [x509_utils_from_pem]: BIO_new failed for certificate
[07:40:21:113] [4481:00001182] [ERROR][com.freerdp.crypto] - [tls_print_certificate_name_mismatch_error]: @@@@



1.3 <u>Build Number of the Machine and the Windows NT version</u>

Question 1; What is the Build Number of the target workstation?

Explanation:

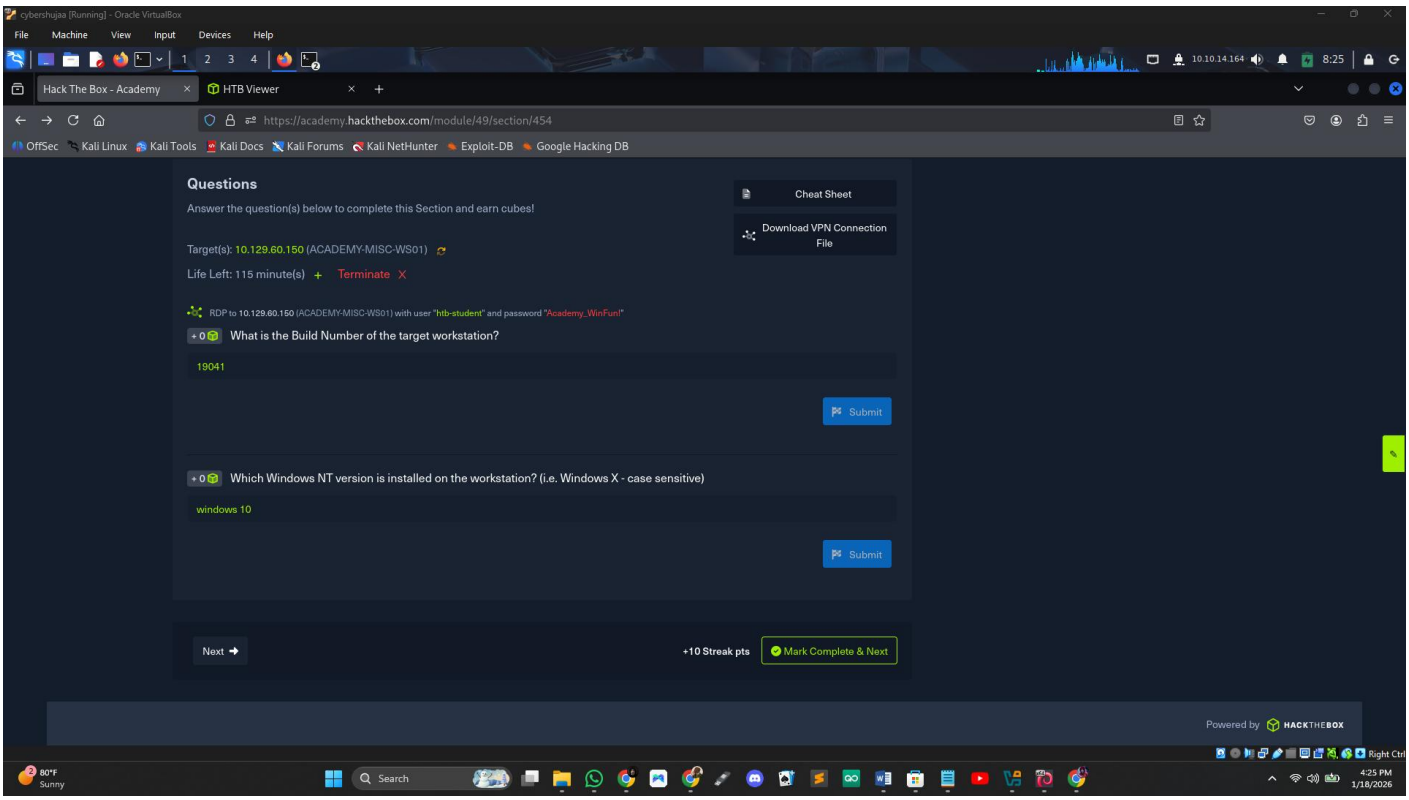
I got to run the target windows and through the use of the command; “Get-WmiObject -Class win32_OperatingSystem | select Version,BuildNumber”, I got to see and to find the build number of the windows target given.

Question 2; Which Windows NT version is installed on the workstation? (i.e. Windows X - case sensitive)

Explanation;

For the Operating system, the interface itself, through previous past interactions with most Windows OS, I noticed the target’s version to be Windows 10 OS.

This is a screenshot of the first 2 questions;



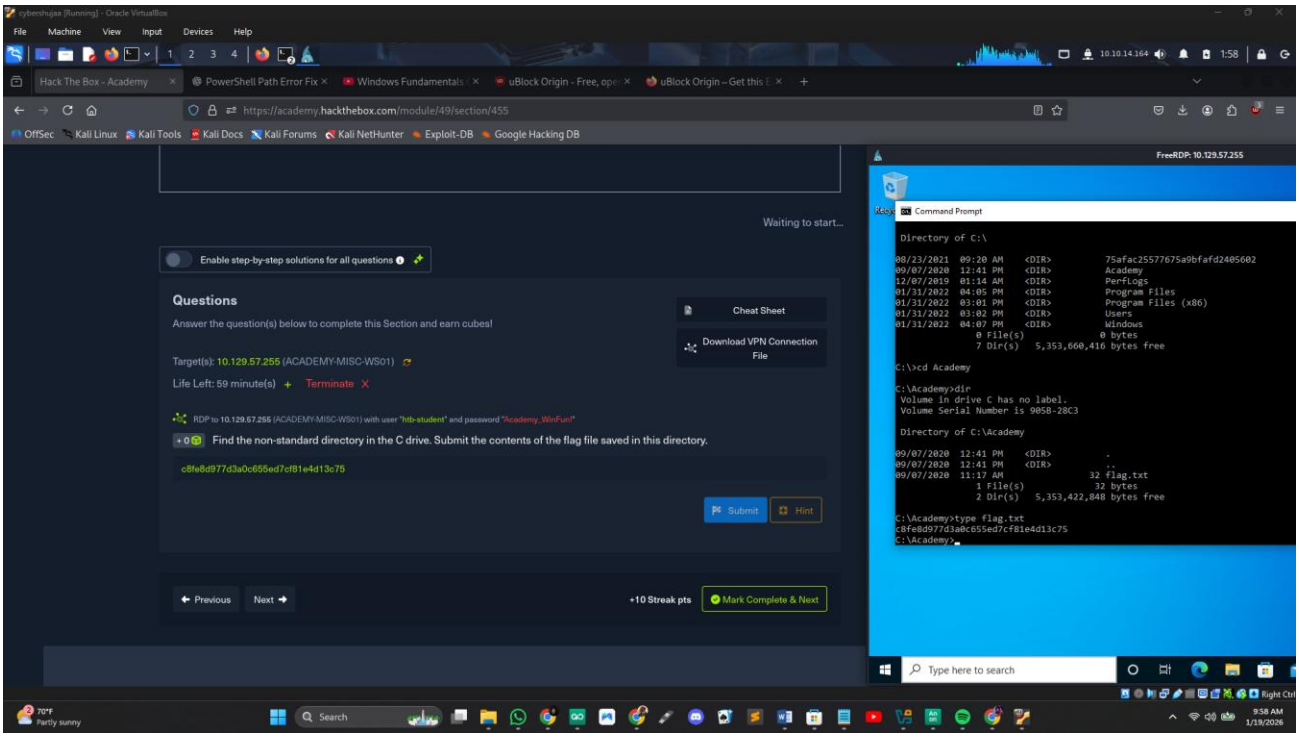
2.2 OPERATING SYSTEM STRUCTURE

I explored the Windows operating system structure by learning how files and directories are organized within the Windows file system. The session talked about the root directory, which is usually on the C:\ disk, and what it does: it is the boot partition where the operating system and core components are put. I learnt what important folders like Program Files, Program Data, Users, and Windows are for and how they keep system files, applications, and user data distinct.

I learned how Windows manages system resources and application data better by doing hands-on command-line exercises with tools like dir and tree that let me see and move around the directory hierarchy. This expertise is useful in real-world systems analysis, troubleshooting, and security assessments.

Question part:

Explanation for the answer in hand/ submitted, the nonstandard was in flag.txt file in the Academy directory (was given as hint in the question), used ‘type flag.txt’ since we are in a windows distro and wanted to see the context in the txt file.



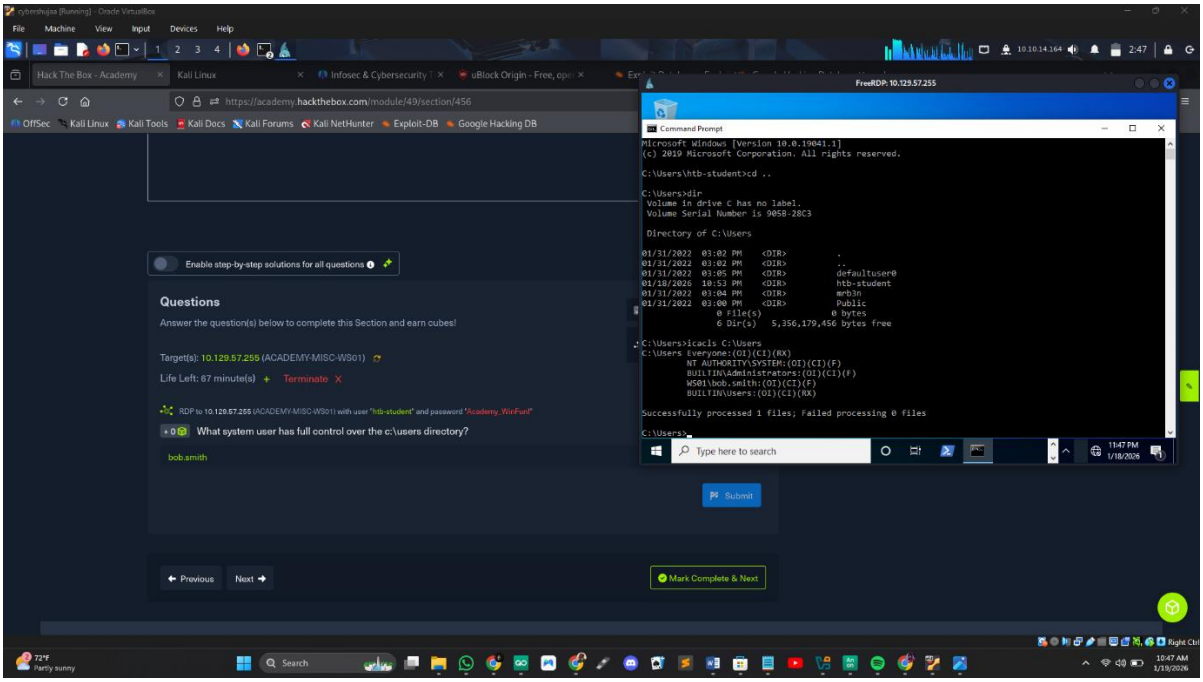
2. FILE SYSTEM

I investigated Windows' file system organization, concentrating on FAT32, exFAT, and NTFS. I discovered that NTFS is more powerful and offers features like file permissions, journaling, and support for very large drives, while FAT32 is compatible with nearly all devices but has limitations, such as not being able to handle files greater than 4GB. Additionally, I had practical experience with NTFS permissions, learning how to control access using the icacls command and how files and folders inherit rights from their parent folders. I gained useful abilities for file management, troubleshooting, and addressing security in real-world systems as well as an understanding of how Windows maintains data security and organization.

Question: ?

Answer explanation, : Used the “icacls” command followed by the directory to whom one want to get information of.

Visual image of my answer:



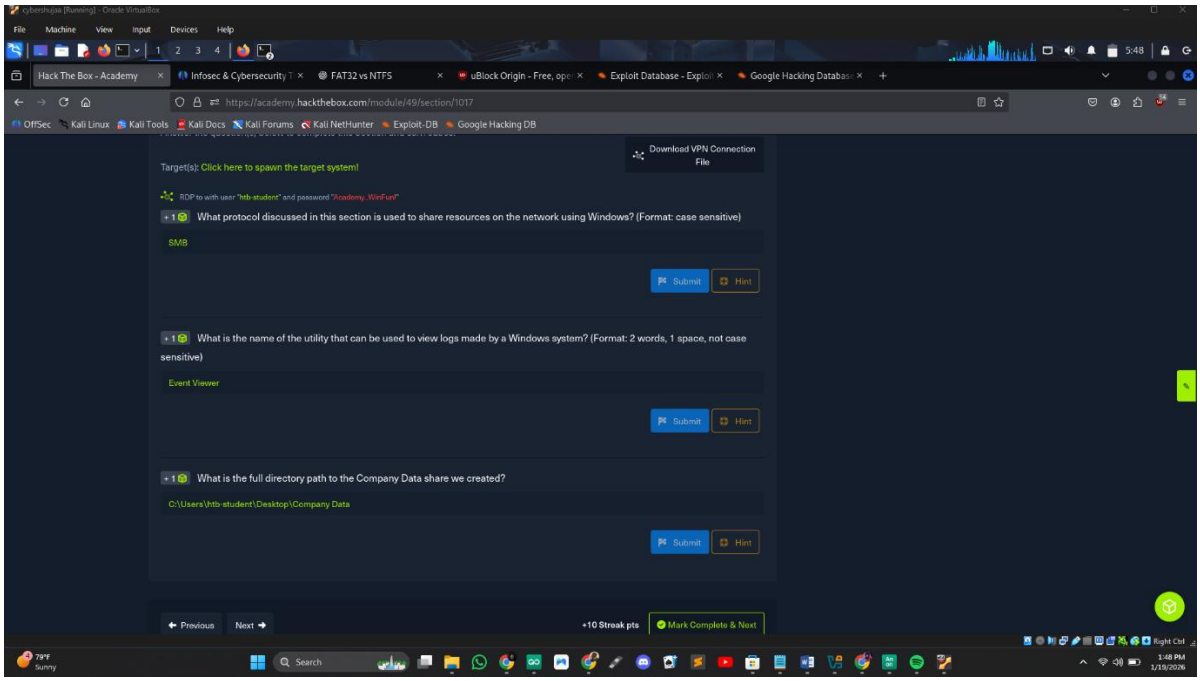
2.1 NTFS VS. SHARE PERMISSIONS

I learned how Windows uses NTFS and share permissions to control access to files and folders. The session explained the differences between local (NTFS) permissions and network (share) permissions, how they are applied, and how the most restrictive permission takes effect when both are used together. This knowledge is essential for securing Windows systems and identifying permission-related security issues.

Questions Part:

Answers Explanation: all of the answers in this module were all based in the notes read in the read section of the learning module

Here is a screenshot of my response:



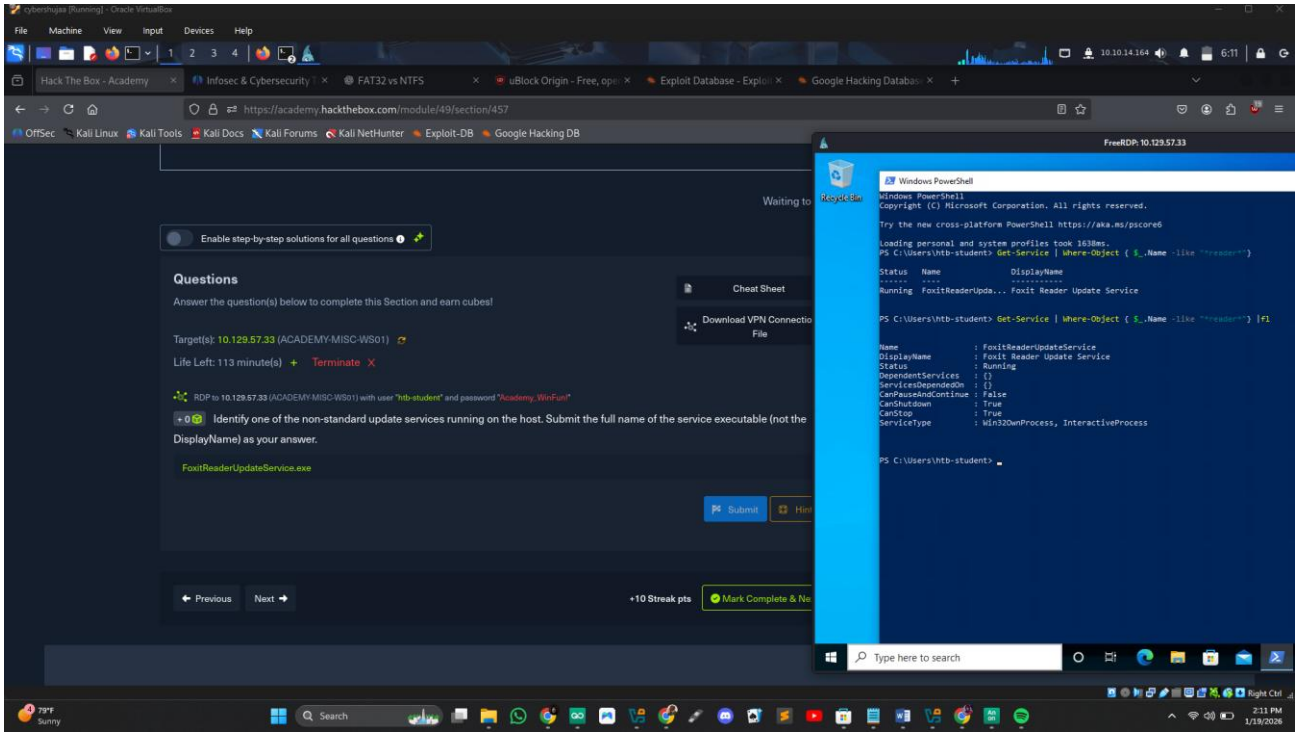
3. WORKING WITH SERVICES AND PROCESSES

In this module, I learned how Windows manages running programs through services and processes. The session covered how to identify, monitor, and interact with active services and processes using built-in tools such as Task Manager and command-line utilities. I also learned how services run in the background, often with elevated privileges, and how misconfigured or vulnerable services can be abused during security assessments. This understanding is important for system troubleshooting and identifying potential attack vectors in Windows environments.

Question part;

Answer explanation: I used an extention to the initial command "…|fl" with this,the full name of the pdf viewing application/ software, will be displayed separately all in a line. Since its Windows OS, I added the '.ex'extention

Visual answer (This one I made sure to add the attacked target since its where the command is visible enough so is the output);



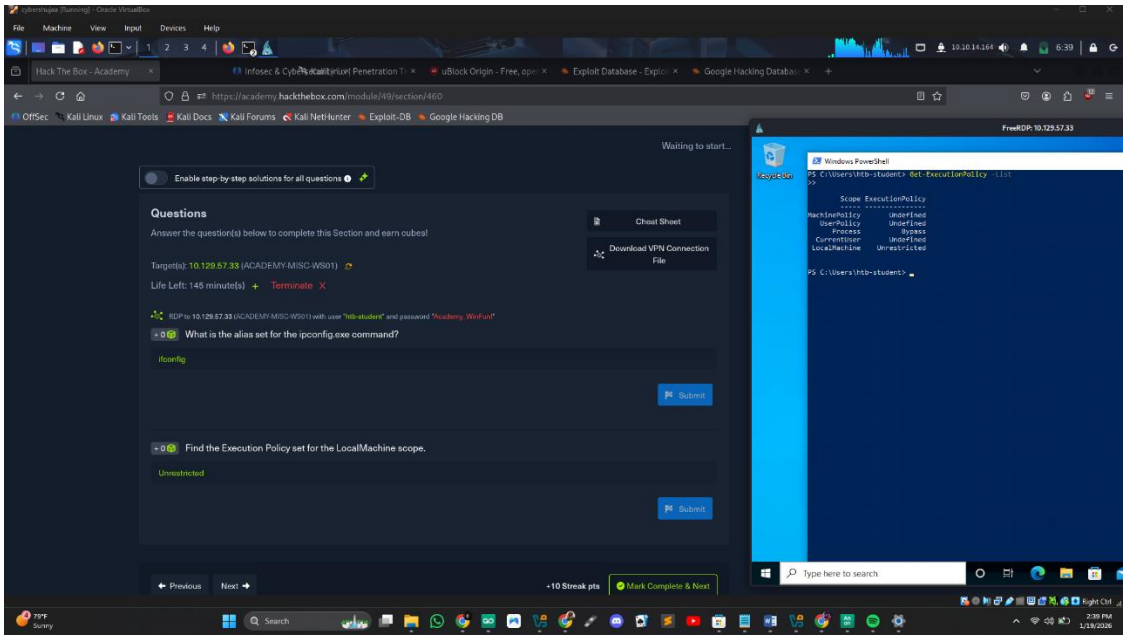
4. INTERACTING WITH WINDOWS

In this section, I learned different ways of interacting with the Windows operating system using both graphical and command-line interfaces. The module introduced the Graphical User Interface (GUI) and explained its importance in making Windows accessible to everyday users and system administrators. I also learned about Remote Desktop Protocol (RDP), which allows users and administrators to access and manage Windows systems remotely through a graphical interface. In addition, the session covered the use of the Windows command line, including Command Prompt and PowerShell, where I practiced running commands, viewing help documentation, and using cmdlets, aliases, and scripts. The exercises also introduced PowerShell execution policies and how they affect script execution. Overall, this section helped me understand how Windows systems are accessed, managed, and automated in real-world and security-focused environments.

Question part;

Answer explanation: The command "Alias" in the powershell helped in answering of the first question. And for the second question, I used a command read on the notes, which showed the policy (execution policy), of the Local machine

Below is the page of my answers as well as the attacked machine used;

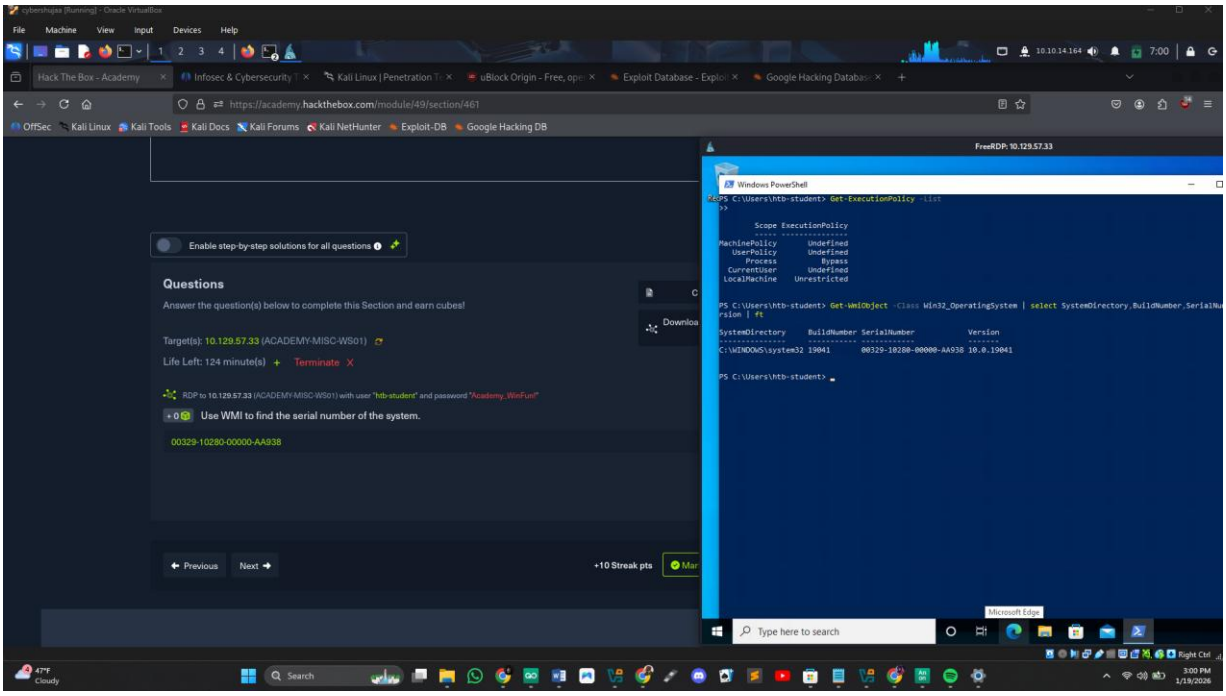


4.2 <u>Windows Management Instrumentation (WMI)</u>

As I covered this module, I learned about Windows Management Instrumentation (WMI) and how it is used to monitor and manage Windows systems. The module explained the role of WMI as a core Windows component that allows administrators to collect system information, manage configurations, and perform actions on both local and remote machines. I practiced using WMI through the command line with WMIC and through PowerShell using modules such as `Get-WmiObject` and `Invoke-WmiMethod`. This section highlighted how WMI can be used for tasks like system enumeration, process management, and remote administration, as well as its importance in both defensive and offensive security operations.

Question Part

Answer Explanation: Using the command: “Get-WmiObject -Class Win32_OperatingSystem | select SystemDirectory,BuildNumber,SerialNumber,Version | ft” will



5.1 Microsoft Management Console (MMC)

In this section, I learned about the Microsoft Management Console (MMC) and its role in managing Windows systems. The module explained how MMC uses snap-ins to group administrative tools for managing hardware, software, and network components from a single interface. I also learned that MMC can be customized to include only the required tools and can be used to manage both local and remote systems, making it a useful utility for system administration and Windows environment management.

5.2 Windows Subsystem for Linux (WSL)

Here I got to know about the Windows Subsystem for Linux (WSL) and how it allows Linux binaries and command-line tools to run natively on Windows systems. The module explained the purpose of WSL, its evolution to include a real Linux kernel in WSL 2, and how it can be installed and accessed through PowerShell. I also learned how to launch a Bash shell, navigate the Linux directory structure, and access Windows files from within WSL. This section demonstrated how WSL enables seamless interaction between Windows and Linux environments, which is useful for development, system administration, and security tasks.

6.1 Desktop Experience vs. Server Core

This section introduced Windows Server Core as a lightweight server installation option focused on efficiency and security. It explained how Server Core operates without a full graphical interface, relying instead on command-line tools, PowerShell, and remote management for configuration and maintenance. The module highlighted its reduced resource consumption and smaller attack surface, as well as its limitations in supporting certain GUI-based applications. Overall, this section emphasized how the choice between Server Core and Desktop Experience depends on operational needs and the administrator's level of expertise.

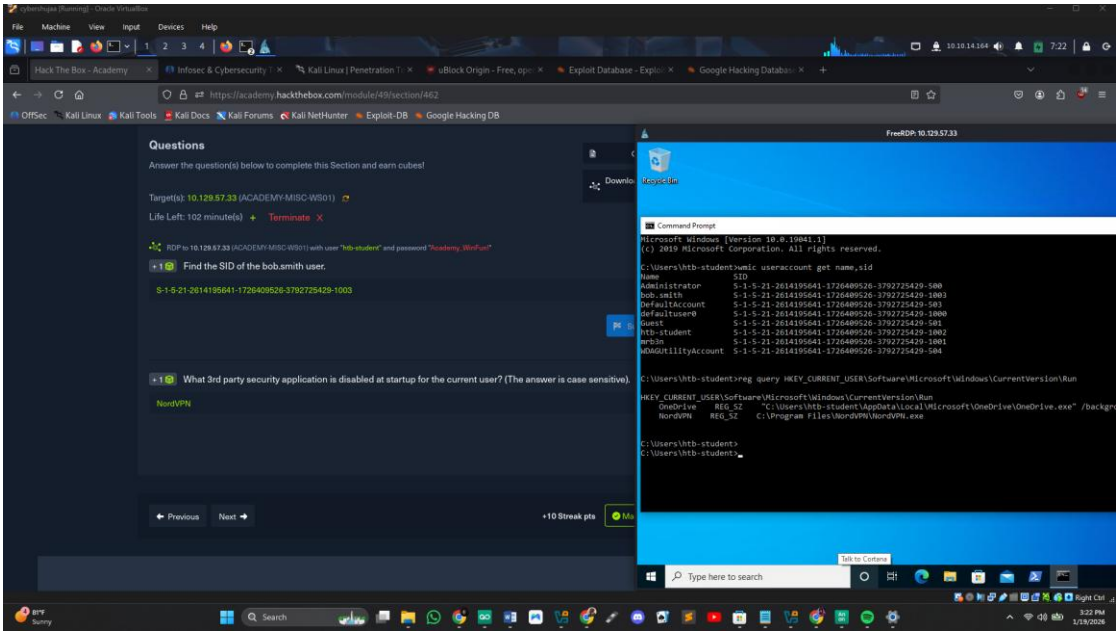
6.2 Windows Security

Digging into Windows internals gave me a clearer picture of how security is enforced across the operating system. I examined how Windows controls access through mechanisms like Security Identifiers (SIDs), access tokens, ACLs, and the Security Accounts Manager (SAM), and how these components work together to authorize users and processes. I also explored the role of User Account Control (UAC) in limiting unauthorized changes and looked closely at the Windows Registry as a central store for system and user configurations. Beyond core concepts, I worked with AppLocker, Group Policy, and Windows Defender Antivirus to understand how administrators harden systems, restrict applications, and detect threats using a layered, defense-in-depth strategy.

Question:

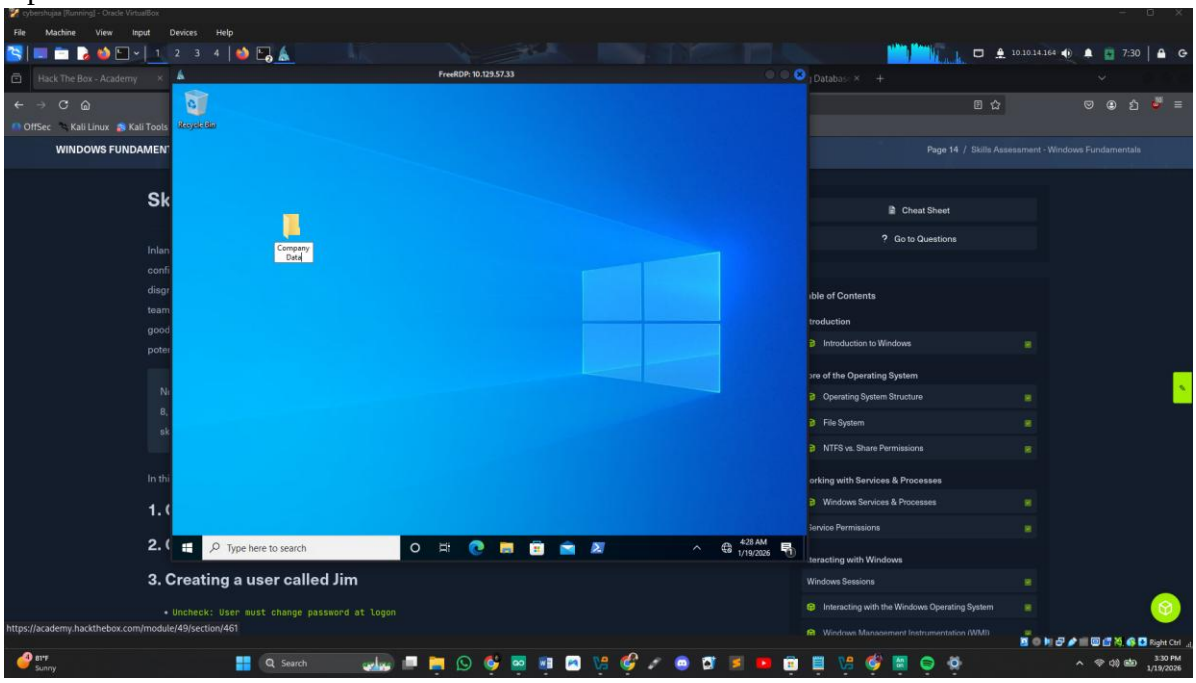
Answer: For the two questions I got to use the cmd to run the commands, as indicated in the target device attached in the screenshot provided below.

Below is the screenshot for my response:

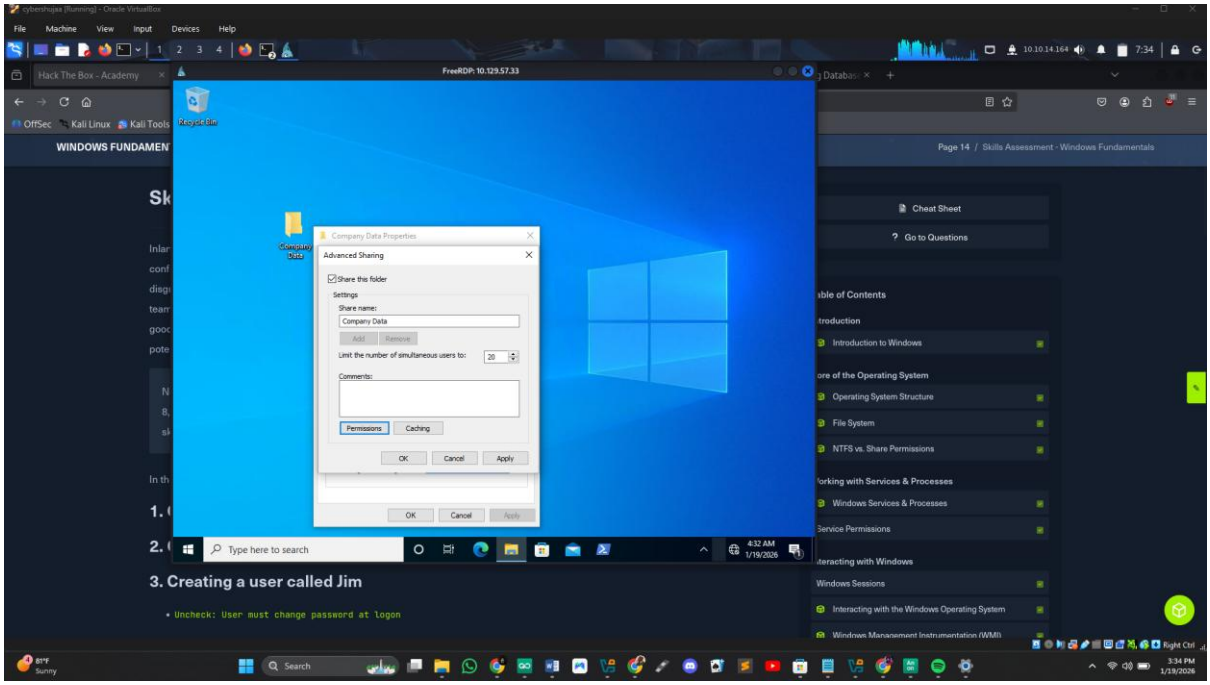


7 SKILLS ASSESSMENT

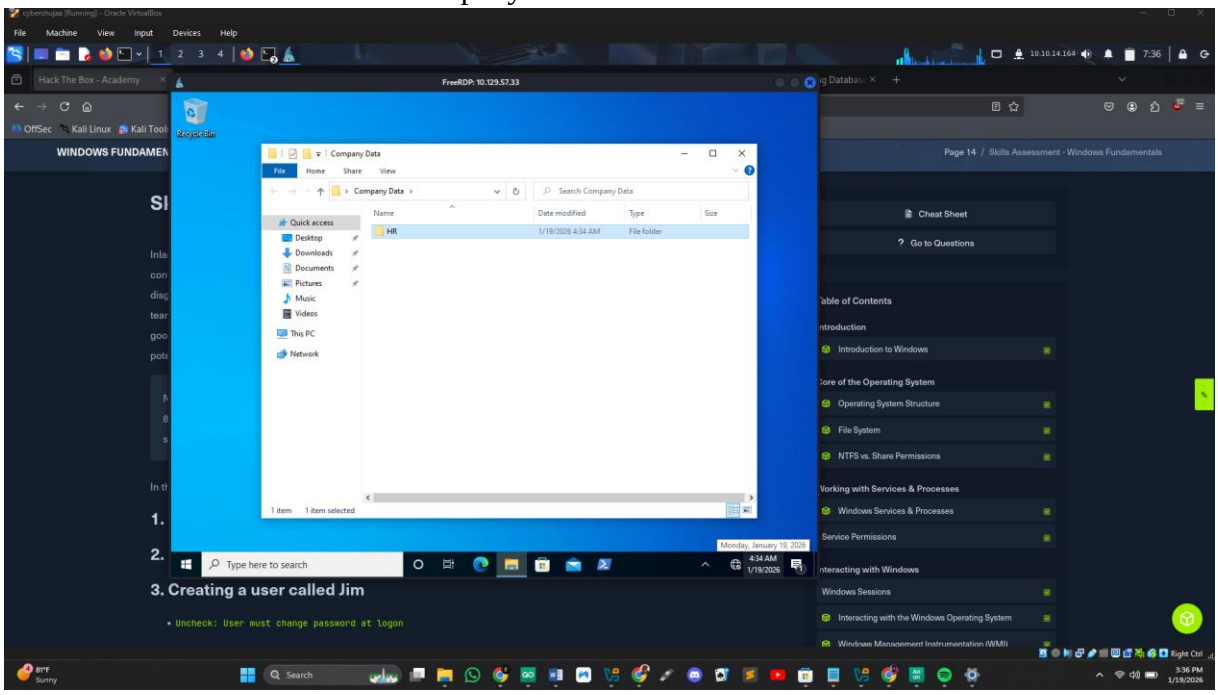
1. I opened my machine, attacked machine
2. Opened and created a new folder



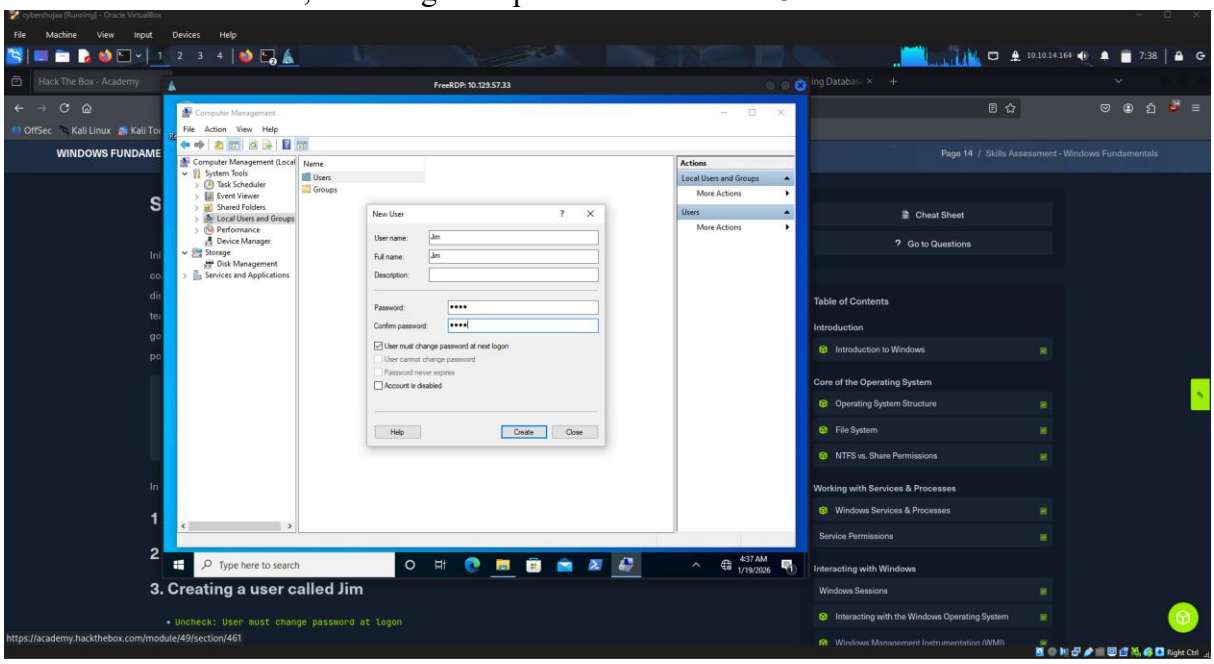
3. Edited the propertied of the folder navigated to sharing tab and changed it so it can be a shared folder



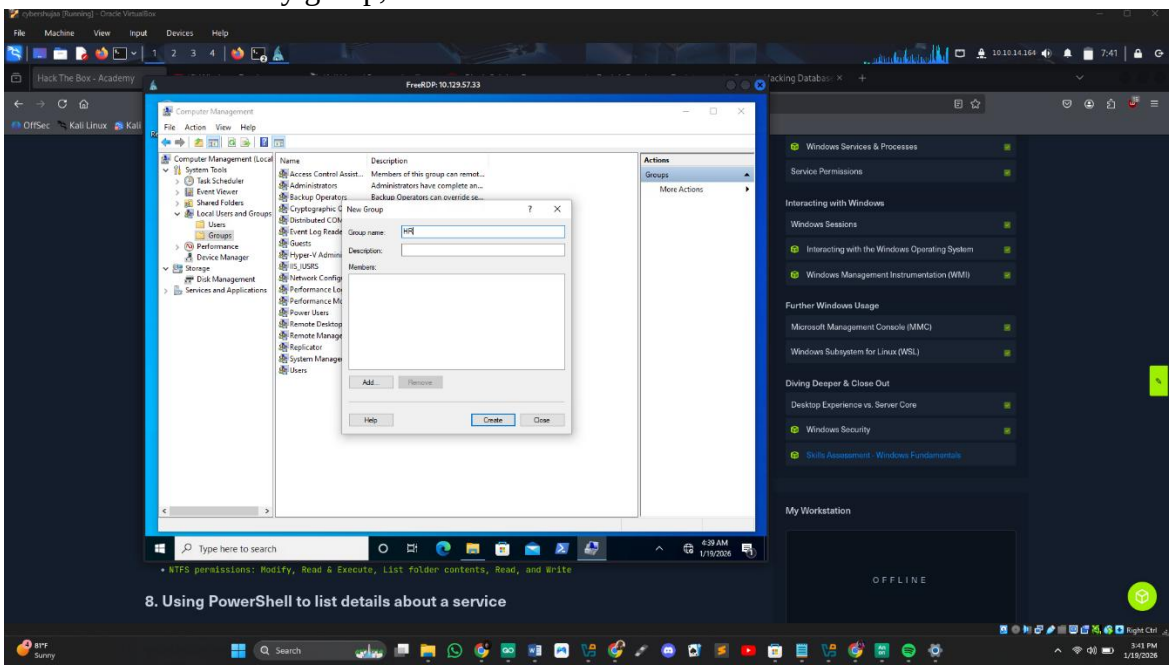
4. Created a HR folder inside our company data folder



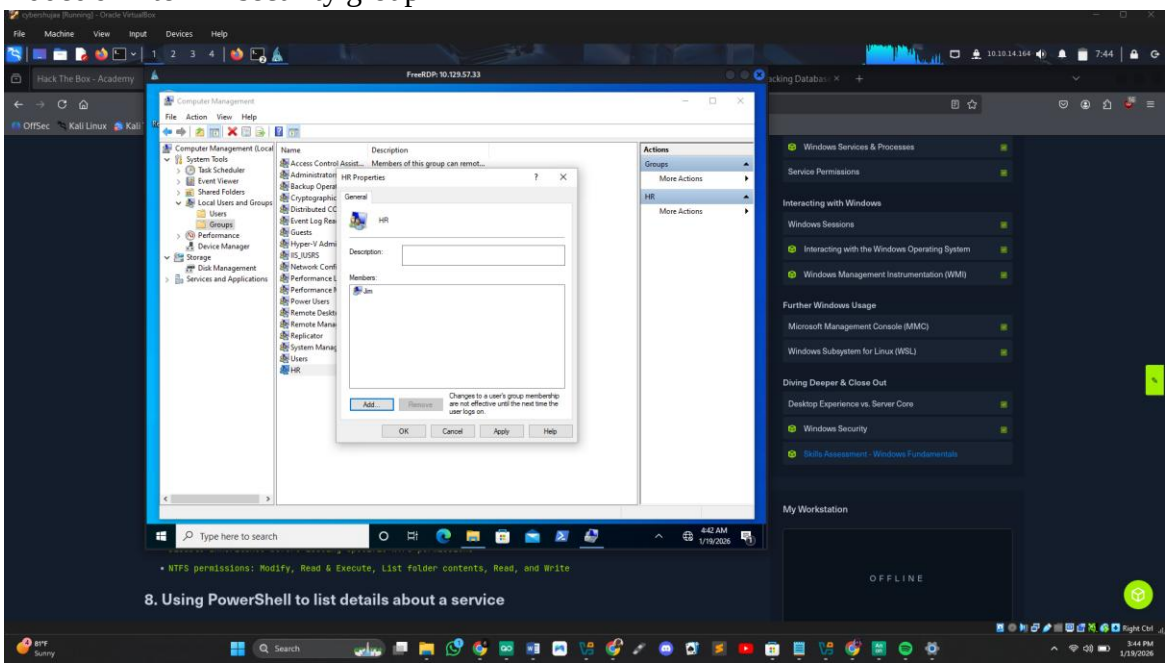
5. Created a new user Jim, and assigned a password to him “1234”



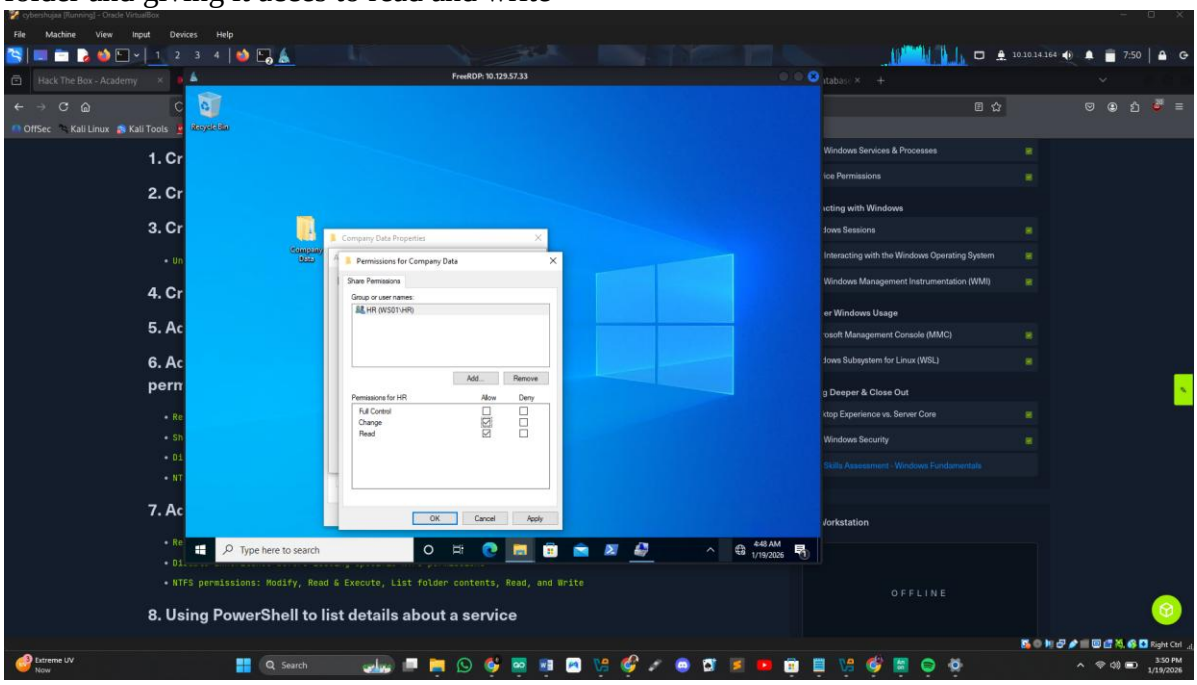
6. Created a new security group, HR



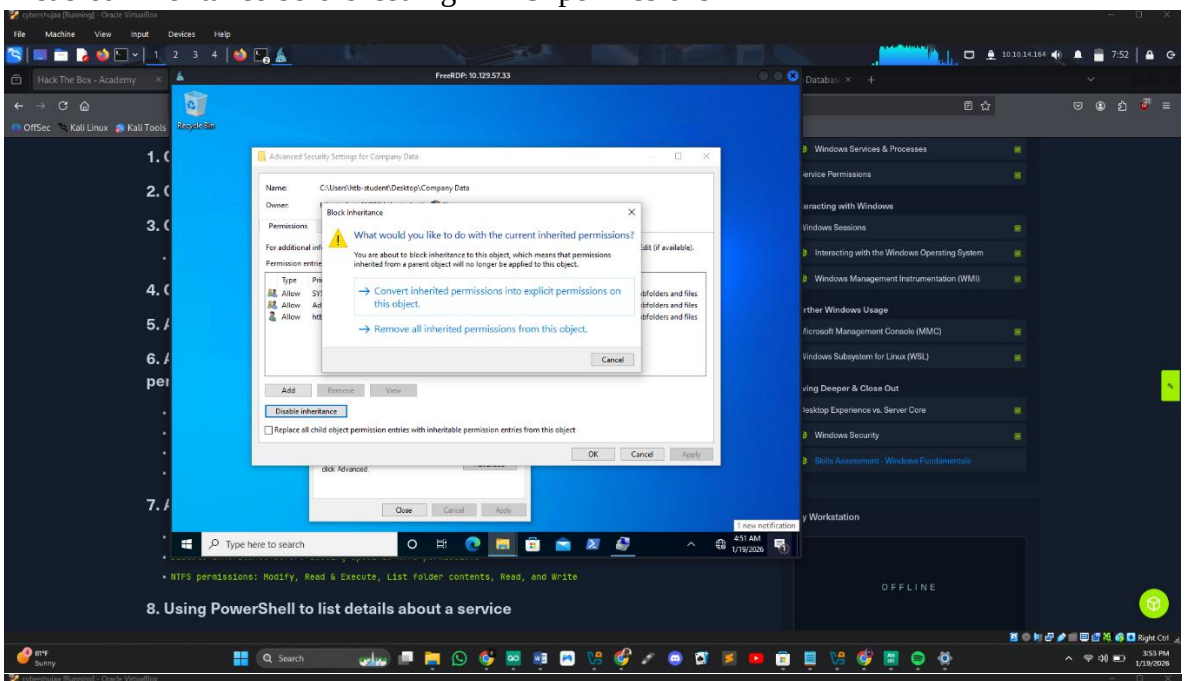
7. Added Jim to HR security group

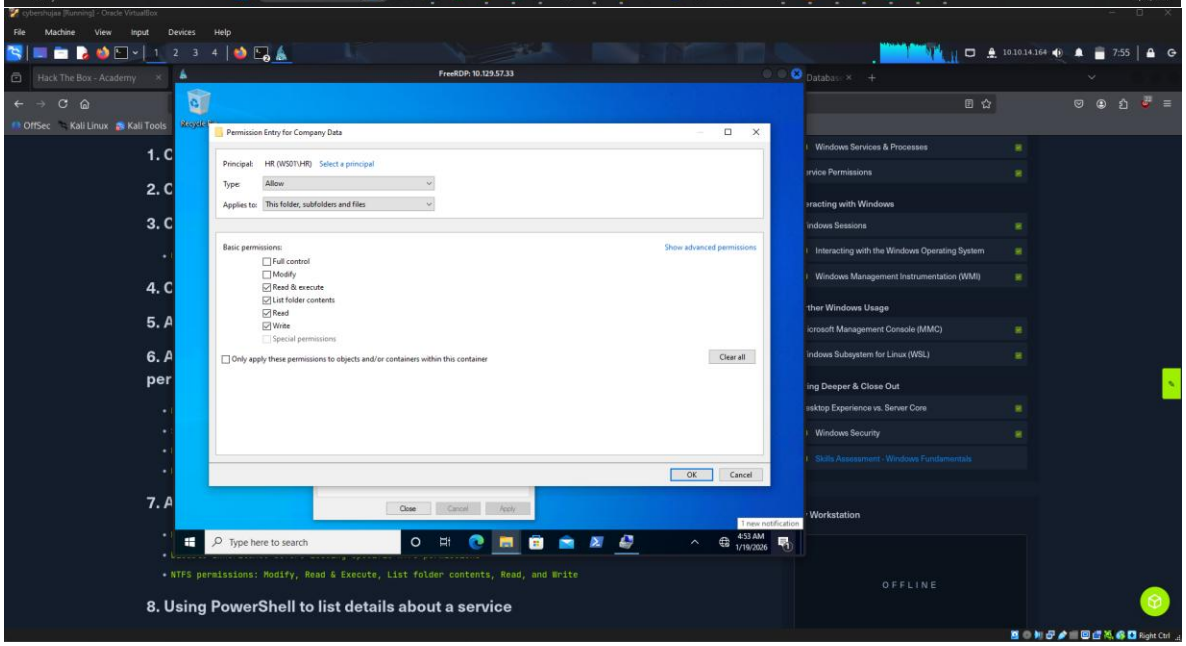


8. Removing the group that was on shared folder, and adding the HR group to shared folder and giving it acces to read and write

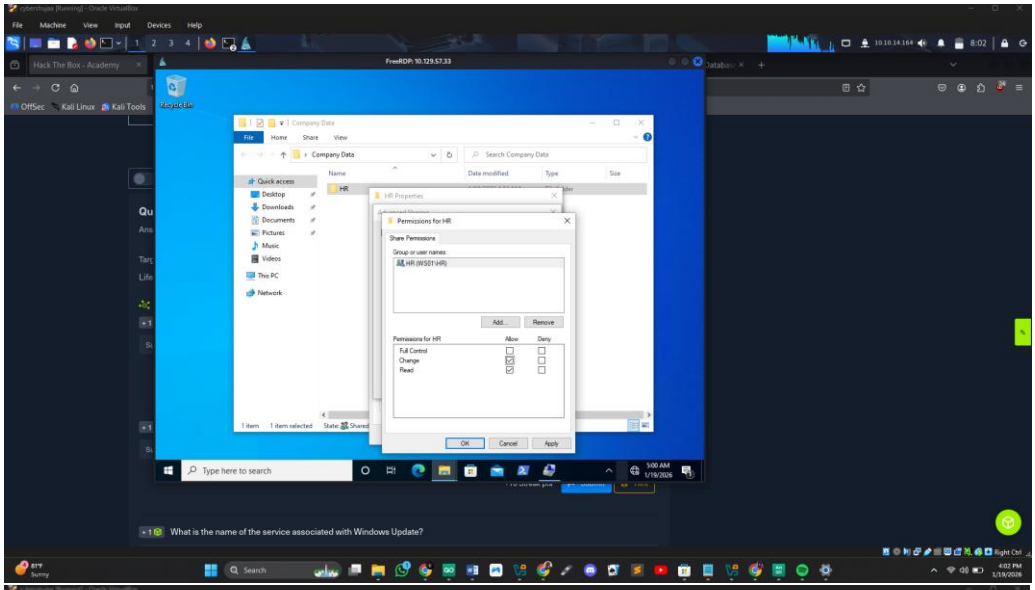


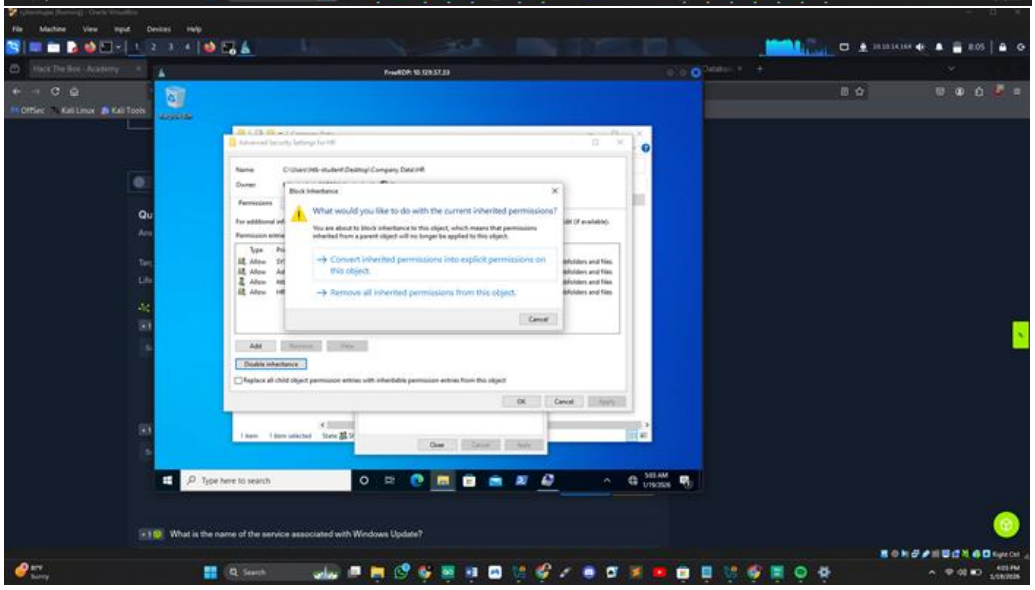
9. Disabled inheritance before issuing NTFS permissions

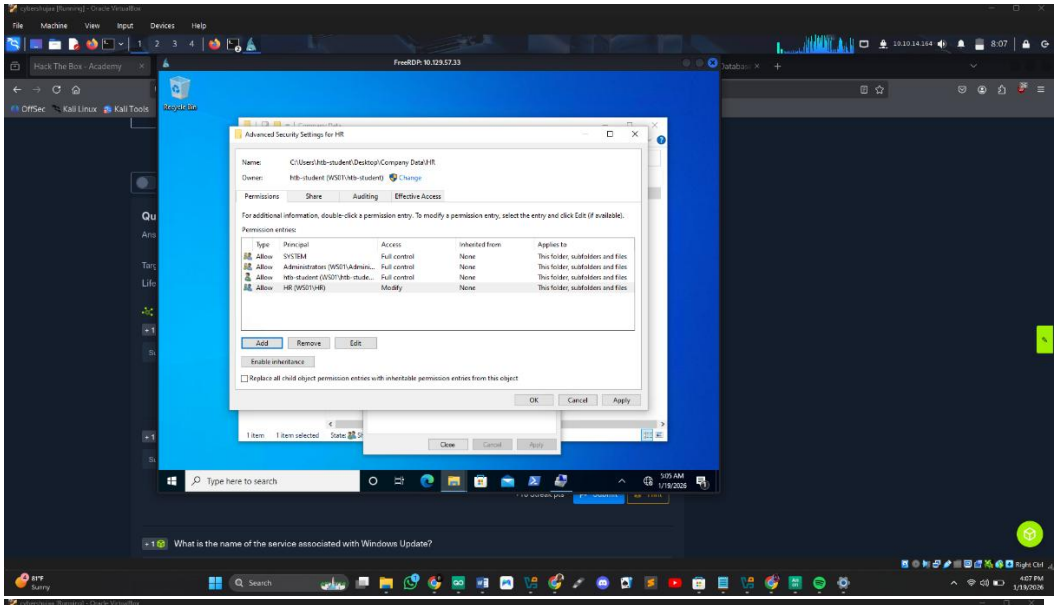


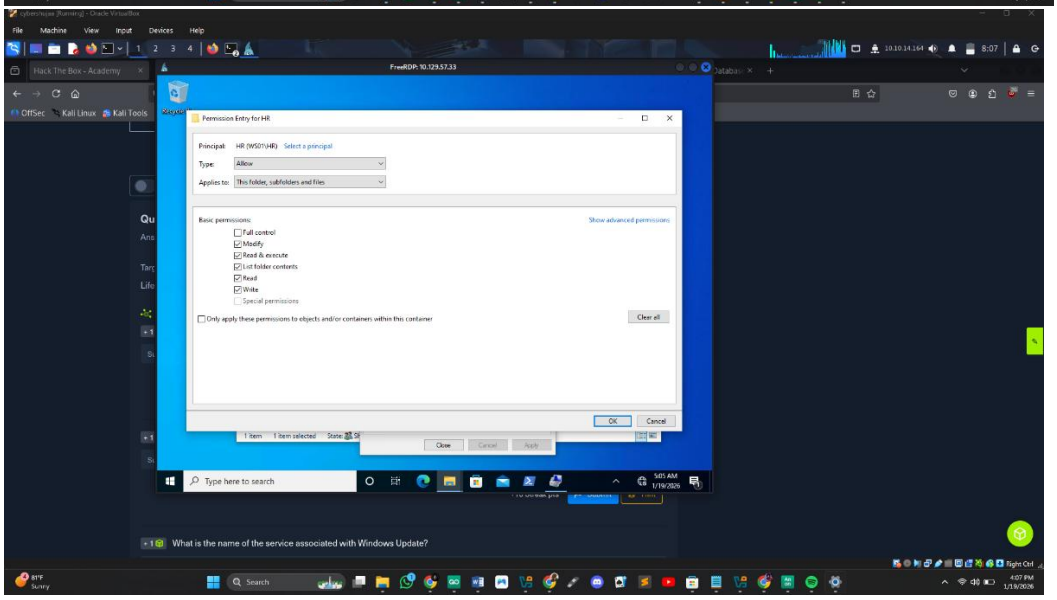


10. Adding HR security froup to the NTFS permission list of the HR subfolder , removing default groups that were present, disabled inheritance and enabled NTFS permissions as stated


Company Data
HR Properties
Permissions for HR
Share Permissions
Group or user names:
HR (WS01\HR)
Add...
Remove
Permissions for HR
Allow
Deny
Full Control
Change
Read
OK
Cancel
Apply
What is the name of the service associated with Windows Update?


Block Inheritance
What would you like to do with the current inherited permissions?
Convert inherited permissions into explicit permissions on this object.
Remove all inherited permissions from this object.
Cancel
Disable inheritance
What is the name of the service associated with Windows Update?





11. Using powershell to list details about a service

Question Part

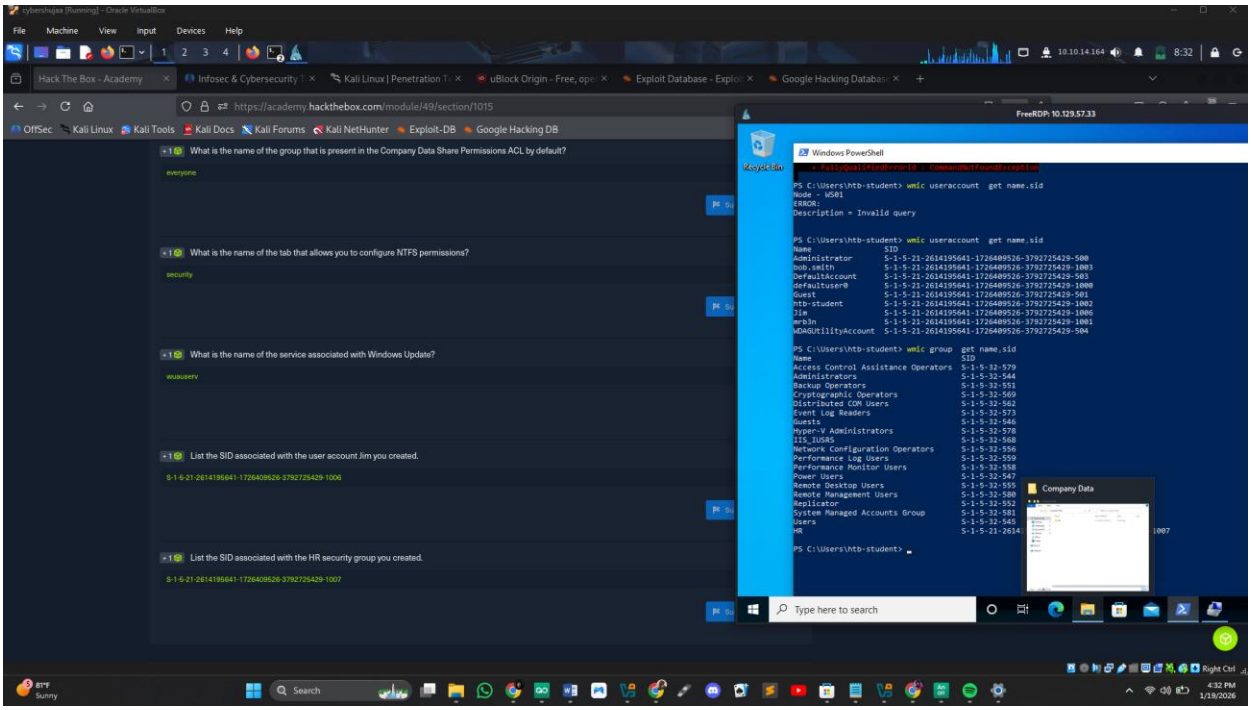
Answers Explanation: for the answers to the questions, the first 2 were direct for someone who was keen in their work all along, so for me they were straight forward.

For the 3rd, I used the command "Get-Services" on powershell of the attacked machine and scrolled till where windows update was so as to find the service associated with windows update

For the last 2, using the command "wmic useraccount get name,sid" and interchanging the key word, useraccount, to groups helps one get SID for every group and username in the machine

Below is a visual image representing the answers responding;

Disclaimer, for the last image the words font was reduced, so as to accumulate all the questions in one image



6 CONCLUSION AND SUMMARY

. Going through the Windows Fundamentals module really helped me understand how Windows works behind the scenes. I got to see how processes, services, and the registry all connect, and how security features like SIDs, UAC, and access control keep the system safe. I also spent time using tools like PowerShell, WMI, MMC, and Windows Defender, which made it easier to see how administrators manage and protect a system. This module gave me a lot of hands-on experience, and I feel much more confident navigating Windows now. I'm excited to keep building on what I've learned and explore more advanced Windows security topics in the future.